

Constructive, Evidence-Gated Generation of Long English Letter-Level Palindromes

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Abstract

We study the constructive generation of long English palindromes under a strict letter-level definition. A candidate must be an exact palindrome, originate from an auditable construction rather than borrowed catalogue text, avoid repeated or self-palindromic units, and survive a blinded human readability study. Our workbench keeps character obligations live while it builds ordinary scenes: typed grammar and semantic slots propose the next words, and targeted repairs are selected from the current mirrored-letter debt. Ten deliberately orthogonal construction lanes share an independent two-pointer and hash verifier, provenance records, and a novelty registry. The current development snapshot contains 5,010 candidates from 294 route summaries; 79 are exact, but none passes the mechanical admission gate and no row has yet been promoted by readers. We report the rendered candidates and the failed gates explicitly, then use each failure to define the next repair operator. The contribution is therefore a reproducible construction and evaluation protocol aimed at a genuinely readable long output, not a speed benchmark or a proxy-only result.

1 Task and acceptance gate

Let x be a rendered English passage. We normalize by lowercasing and retaining ASCII letters only,

$$n(x) = \text{letters}(\text{lower}(x)), \quad (1)$$

and require $n(x) = \text{rev}(n(x))$. Punctuation and spacing are displayed for readers but cannot change the equation. The target is an intact, multi-clause passage of at least 100 normalized letters; the familiar 38-letter seed is used only as a development benchmark.

The hard gate has four independent parts. First, a standard-library two-pointer check and a separately implemented forward/reverse hash check

must agree. Second, the provenance ledger must show that the text was constructed by the current method, with no catalogue import, scaffold relexicalization, repeated multiword unit, word-order reflection, or self-palindromic word/span. Third, the rendering must be ordinary prose with distinct lexical choices, not a list or fragment. Finally, blinded readers must judge it readable and coherent. Programmatic language scores can rank or diagnose candidates; they never certify the last condition.

2 A working construction loop

The constructor maintains a state $(L, R, d, \sigma_L, \sigma_R, \rho)$, where L and R are the committed left and right tapes, d is the vector of mirrored character debt, σ stores grammar and agreement state, and ρ is the semantic scene graph. At each step it (i) chooses a typed scene slot or boundary, (ii) proposes a short ordinary word or inflection compatible with σ and ρ , (iii) updates the opposing character obligation immediately, and (iv) retains the best frontier by length and unresolved debt. A repair may replace a complete semantic slot, resegment an exact tape, change a dependency seam, or alter an inflection/clitic boundary. A run is not counted as progress when it merely resamples the same lane or duplicates a prior signature.

This is a character-constrained generation problem in the spirit of prior palindrome enumeration and lexically constrained decoding (Papadopoulos et al., 2015; Hokamp and Liu, 2017; Lu et al., 2022), but the character obligation is coupled to a live scene and grammar rather than applied after a sentence has been written.

The ten lanes are intentionally different views of this same loop: character-level LM decoding; exact-tape grammatical resegmentation; dependency-tree seam CSP; agreement-carrying morphology; CFG/Earley character intersection;

a human-authored scene lattice with live equations; semantic valency and attachment; inflectional and clitic boundary search; a flat, compositional grammar without nested palindrome spans; and exact-candidate repair by semantic slot substitution. Each lane writes its rendered text, normalized tape, independent checks, source hashes, novelty result, and a concrete next repair to a JSON artifact.

The implementation also exposes an explicit totality witness for arbitrary requested lengths. This witness is useful for testing the state machine but is tagged as non-reader output and cannot satisfy the admission gate. Thus termination of the software is never confused with success of the linguistic task.

3 Current rendered evidence

Table 1 gives one fresh surface for each lane family. These are actual outputs, not templates; lengths count normalized letters. Exactness and readability status are shown together so that a high-scoring but invalid surface cannot become a headline result.

The aggregate also records a 1,922-letter recursive clause-growth diagnostic; it is retained as a scalability probe, but fails exactness and the repetition and length gates, so it is not a reader-facing result. Among the complete prose frontiers, the CFG/Earley lane reaches 270 letters and the character-LM lane reaches 227 letters; neither closes exactly. A tape resegmentation lane also found the 21-letter catalogue control “a man a plan a canal panama”; its exactness is independently confirmed, but catalogue provenance excludes it. These controls are useful for checking the gates, never for inflating the result set.

The latest typed-frame repair adds the fresh 40-letter sentence “The baker carries a letter near the quiet harbor.” Its eight live obligations remain open, while every mechanical quality check other than exactness passes. The next run changes only the adjunct boundary lexeme selected by the first residual obligation. That targeted repair changes “near” to “by”, yielding “The baker carries a letter by the quiet harbor”; it remains non-exact at 38 letters, below the mechanical length floor, and records the determiner slot as the next repair.

Two further joint searches extend the constructive evidence. A central-pivot CSP retains the intact 113-letter pair “The archivist records the

carefully folded winter map at dawn. A patient sailor observes a weathered cedar rescue boat beside the river.” Its first mirrored characters already disagree, so both independent audits reject it. A semantic-slot lattice renders 108 jointly selected states and prunes all 108 before admission. A bidirectional decoder also reproduces the 51-letter “Doc, note: I dissent.” palindrome, but the provenance gate identifies it as catalogue text and keeps it out of the result set.

The word-pair graph contributes a 164-letter intact scene frontier but no closure. Paired semantic mutation produces two fresh 61-letter controls and no closure. Finally, weighted reverse-tape segmentation retains two authored 103–104-letter clauses but finds no legal segmentation. These runs are separate construction states; each leaves an explicit repair rather than being counted as a larger duplicate sweep.

A residual repair then produces a 239-letter four-clause scene from fresh independent clause authoring; its first boundary remains unmatched under both audits. A separate semantic-slot repair adds two new 108–109-letter scenes, again with complete slots and no exact closure. Both are retained as the next reader-facing controls and specify a local boundary or paired-slot repair.

The character-LM grammar decoder retains a 77-letter legal-prose frontier. A new dependency-seam CSP adds a 105-letter typed scene, and an agreement-carrying morphology/clitic transducer adds a 122-letter scene. All three are fresh, ordinary-order constructions with independent audits and explicit repairs; none closes the global character equation.

The joint CFG/Earley intersection adds one 140-letter scene from a paired bidirectional chart. A 729-row cross-product variant is explicitly excluded as a duplicate sweep, preserving its evidence without counting it as progress. The rendered controls are “a patient child carries the river trail at dusk. the keeper opens old wooden gates before dawn,” “The patient archivist carefully maps a quiet museum archive before dawn. Curious visitors study faded stars beside the river,” “When the careful archivist restores the damaged ledger, she records each witness name and sends the sealed copy to the harbor office before winter,” and “The patient gardener waters the cedar seedlings beside the schoolhouse before sunrise The teacher labels every seedling and stores the tools beneath the quiet porch.” Their independent

Lane family	Rendered surface	Letters	Exact	Next repair
LM + tape	the careful teacher guides a young visitor; a patient sailor carries the red lantern	70	no	held-out LM boundary proposals
Dependency + morphology + CFG	Ava saw radar level civic; civic level radar was Ava	42	yes [†]	typed boundary resegmentation / fresh frame
Scene lattice	At dawn, the patient guide carries a red map to the waiting child.	52	no	replace one typed object slot
Valency / attachment	By noon, the careful nurse labels the sealed vial for the quiet ward.	55	no	preserve valency while changing object/attachment
Inflection / clitic	After rain, the young keeper opens the old gate beside the garden.	53	no	held-out inflection-boundary repair
Flat composition / slot repair	At dawn, the patient guide carries a blue chart to the waiting child.	55	no	satisfy the global character obligation

Table 1: Representative outputs in the 2026-09-16 ledger. [†]The 42-letter seam is exact under both independent checks but is rejected because it repeats self-palindromic words, mirrors word order, and fails the distinct word gate. The other rows are fresh ordinary prose and are retained as repairable frontiers, not as palindrome claims.

audits all reject exactness; each has a named local repair and remains reader-ineligible.

The hand-authored clause lattice then aligns five fresh complete clauses across word boundaries, with an explicit all-different content-word gate. It yields the longest current intact-prose frontier (308 letters): “Mara carries warm bread to the river. The careful pilot studies cloud maps. Children gather bright shells by moonlight. A gardener shelters young cedar shoots. Old friends share stories beside fire. By the hearth, new tales begin. Near the grove, small finches settle. At twilight, the patient tide returns. Beyond the hills, a quiet engine waits. At the shore, Owen listens for bells.” The independent pointer and forward/reverse SHA checks disagree at the first character, so it is explicitly non-exact and non-admitted. Its provenance, fresh lexical bank, and next semordnilap-compatible slot repair are recorded in the run artifact; it is a reader-facing control, not a palindrome claim.

Three follow-up constructions add new state representations rather than larger sweeps. A boundary-conditioned finite-state resegmentation produces a 270-letter six-clause scene; a dependency-boundary CSP exposes 127 live character equations in a marine-biologist scene; and a residual semantic-slot lattice produces a 122-letter observatory scene while locking valency, agreement, and content-word uniqueness. Each has a first-mismatch pointer trace, unequal forward/reverse hashes, fresh provenance, novelty preflight, and a targeted next repair. None is exact or mechanically admitted.

The outside-in scene grammar was then repaired to use distinct held-out right frames instead of reflecting a left clause. Its longest pair is “At first light, the marine biologist records pa-

tient observations beside the sheltered tide pool. At dusk, the coastal engineer maps hidden channels.” (124 letters); six fresh 102–124-letter prose pairs survive an all-different content-word filter. An exact-tape-first control does close on “Able was I; I saw Elba,” but novelty and length gates exclude that known 16-letter catalogue sentence; a fresh 72-letter exact tape is also excluded because its reflected half is gibberish. Finally, a ten-clause residual solver retains a 355-letter clinic scene: “At first light Mara opens the clinic, checks the quiet generators, greets the two nurses, and records the medicine count. She carries clean water to the waiting room, labels each parcel, phones the mountain driver, updates the weather board, thanks the volunteers, and closes the ledger before dusk. After supper she inventories the blankets, answers the radio, repairs a torn notice, and leaves clear instructions for the morning shift.” All three results carry independent audits, provenance, and novelty records; none is mechanically admitted. A finite clause automaton adds a 109-letter archivist/visitor scene, a typed semordnilap lane adds a 100-letter baker near miss beside the rejected short “Stressed desserts.” control, and a two-sided discourse equation adds a 156-letter Nora/harbor scene. These lanes retain dual audits and targeted repairs but no mechanically admitted output.

The fresh ten-lane rerun keeps the construction dimensions separate. The character-LM beam emits a 227-letter archival scene; immutable-tape boundary DP retains a 75-letter prose fallback after finding no grammatical segmentation; dependency seam CSP, agreement morphology, CFG/Earley intersection, and a live human-authored scene lattice emit 123-, 155-, 270-, and 117-letter scenes; typed valency/attachment and

semantic slot repair emit 100 and 170 letters; and flat compositional grammar reaches a 208-letter action sequence without nested palindrome spans. Each row is rendered and independently pointer/hash audited with provenance, novelty preflight, and a named repair. A 126-letter exact surface from the inflection/clitic lane is quarantined because it repeats the catalogue clause “A man, a plan, a canal, Panama.” six times; it is not counted as generated readable evidence.

The subsequent held-out inflection/clitic repair replaces that repeated unit with a distinct 128-letter harbor scene carrying plural, past-tense, and possessive-clitic registers. Its pointer and hash audits still reject exact closure, but the route demonstrates a concrete anti-shortcut repair rather than accepting the duplicate control.

Three post-audit continuations extend the construction state without widening a duplicate sweep: a free-center semantic machine (138 letters), an independent phrase-level reverse parser (nine 116–118-letter probes), and a finite-domain clause-equation SAT model (116 letters). Their rendered clauses, independent pointer/hash audits, provenance, novelty preflight, and named repairs are included in the evidence ledger; none reaches the reader gate.

The following dialogue speech-act, typed morpheme-compound, and first-token equation probes add 130-, 68-, and 91-letter ordinary-language renderings. They remain diagnostic controls because the first two do not close and the morpheme realization also fails the lexicon gate.

An A* typed word-boundary probe retains a 140-letter garden/schoolhouse scene, while a single-slot human semantic search retains seven deduplicated 92–97-letter scenes. An attested phrase-bridge preflight found no complete clause and is excluded. These outcomes are recorded as bounded construction evidence, not as larger duplicate sweeps.

Three further Luna continuations add distinct constructive states: a paired lexical grammar renders six fresh 75–82-letter SVO pairs, a free-center semantic bridge renders a 109-letter scene, and a productive grammar-pair composition grows complete paired productions through 55, 114, 168, and 223-letter states. Each has independent pointer/SHA validation, provenance, novelty preflight, and a first-residual repair; all remain non-exact and are retained as bounded evidence rather than duplicate sweeps.

A typed onset/rime grapheme constructor contributes 12 corrected 37–47-letter complete clauses, and a reader-first function/inflection editor contributes five 96–104-letter scenes. Both remain non-exact after independent audits. An entailment-preserving active/passive rewrite is retained only as excluded failure evidence because novelty preflight found overlap with existing voice and information-structure families.

Three additional Luna lanes keep the construction state distinct rather than resampling a prior sweep. A constrained edit program renders four intact 97–106-letter versions of one cartographer/ranger scene while lowering mirrored-character debt from 51 to 44; an append-algebra probe renders four complete 28–102-letter clause-growth states and records the suffix-obligation invariant; and a discourse-relation involution renders 12 complete 43–56-letter cause/effect or contrast propositions. Each has fresh provenance, a novelty preflight, independent pointer/SHA replay, and a named next repair. None is exact or reader-eligible.

The next repair wave adds three more orthogonal states. A typed semantic center-out solver renders 12 complete 95–109-letter scenes; a lexical word-equation intersection renders four complete 133–143-letter scenes; and a joint slot/boundary repair renders nine complete 45–62-letter scenes. The representative surfaces are “The patient curator shelters the fragile maps during the storm; the careful teacher copies the final field notes beside the window.”, “The curator labels the fragile map before the archivist stores the ledger, while rain gathers softly against the western windows and visitors wait beside the reading room.”, and “The quiet curator labels the old faded map before dawn.” Their normalized tapes are independently rejected by two-pointer and forward/reverse SHA checks; each run records provenance, novelty, and a concrete boundary repair.

Two further Luna lanes keep the search constructive. A character-level beam over typed semantic frames emits “The young botanist studies the silver seed cases beside the greenhouse; a careful pilot marks the distant landing lights through the mist.” (115 letters; first mismatch 1; normalized SHA prefixes “e57ceba...” and “99cbaec3...”). A fresh CFG/Earley character intersection emits “At first light, the surveyor records the river current while the baker warms bread for the waiting crew, and the har-

bor keeper checks the lamps before opening the gate.” (137 letters; first mismatch 0; SHA prefixes “a24dca02...” and “8b577593...”). Both are complete prose with independent pointer/SHA checks, provenance, novelty preflight, and a typed next repair; neither is exact or reader-eligible.

Three further Luna constructions are retained as fresh states. A paired semantic CFG produces a 108-letter two-clause scene; a joint semantic/ inflection word-boundary DP produces a 109-letter scene with independent sentence boundaries; and an online clause-order lattice produces a 151-letter four-sentence archive scene. Their rendered surfaces, pointer/SHA audits, provenance, novelty checks, and next repairs are recorded in the evidence ledger. None is exact or reader-eligible.

The newest bounded continuations change the state representation again. An endpoint-aware bilateral seam reserves lexical terminals before interior character decoding (six 100–105-letter scenes); a fresh-scene tape/CFG intersection keeps boundaries mutable and records a 126-letter lexical repair; and a typed semantic-slot neighborhood changes one role-compatible slot in six 109–160-letter scenes. All retain intact prose, independent two-pointer and forward/reverse SHA audits, provenance, novelty preflight, and a concrete next repair. None closes exactly, so none is promoted to the reader packet.

The latest constructive continuations add a nested-free slot-boundary DP, a held-out semantic phrase-edge joiner, a coupled object/attachment repair, a role-typed reversible-clause product, and a POS-filtered reversible-clause composer with appendable frame growth. They produce intact 98–124-letter scenes (including a 111-letter harbor/reader clause), keep character equations live during construction, and record independent audits, provenance, novelty, and the next repair. They add construction states rather than duplicate sweeps; none has yet closed the exact tape.

Two additional fresh states extend this evidence without changing the gate. A seed-free authored-frame insertion lane renders the 83-letter scene “At dawn, the archivist opens the cedar cabinet and records the harbor map, before dusk.” An interrogative/relative-template solver renders the 87-letter scene “Was the quiet curator sure that the young pilot had seen the chart I filed beside the harbor ledger?” Both are complete prose with online character obligations, independent pointer/hash replays, provenance, novelty preflight, and a named

repair; neither is exact or reader-eligible.

The newest Luna queue adds three more non-overlapping constructive states. A live mirrored-tape terminal decoder emits six complete 88–103-letter scenes; a human-authored attachment CSP emits two 105–107-letter museum scenes; and an unbounded bilateral semantic-growth grammar emits three non-repeating clause-growth states up to 140 letters. Each row is independently checked, provenance-backed, and paired with a concrete first-residual repair. None is an exact, mechanically admissible palindrome, so none is represented as a reader result.

The following queue pass adds three boundary-aware states: an independently resegmented reflected-tape grammar (two 103–107-letter scenes), a semantic scene equation lattice with asymmetric token counts (two 111–112-letter scenes), and a finite-state semantic boundary-macro grammar (three growth states up to 74 letters). These are fresh complete prose rows with independent pointer/hash replay, provenance, novelty preflight, and concrete residual repairs; none closes exactly or enters the reader packet.

The reset wave then changes the construction state rather than expanding a prior sweep. A live CFG/Earley character chart renders six fresh scenes up to 102 letters; a held-out cross-word seam CSP renders sixteen 62–75-letter role-typed scenes without a fixed tape; and POS-shaped seam mining feeds fresh authoring of two connected 100–101-letter valency scenes. Each row is actual prose with independent pointer/SHA validation, provenance, novelty preflight, anti-shortcut checks, and a concrete next repair. None closes exactly.

For audit transparency, the corrected grapheme lane’s representative rendering is “The baker repairs the gate; the pilot maps a gate.” (39 letters, first mismatch at 0), while the longest function-edit rendering is “At dusk, Mara carries the blue lantern across the quiet bridge, and Jonah records each rescued name so the town archive can open.” (104 letters, first mismatch at 0). Their normalized forward/reverse digests disagree; neither is presented as a readable palindrome.

4 Independent verification and provenance

For every row, the verifier recomputes normalization, length, first mismatch, two-pointer equality,

and a forward/reverse digest without importing the generator’s booleans. The mechanical admission vector records ASCII support, lexical form, distinctness, catalogue-family exclusion, repeated-unit and self-palindrome exclusion, and non-word-order symmetry. The registry assigns one signature to each run and rejects duplicate families before aggregation. On the authoritative snapshot the registry contains 395 unique artifacts (32 pre-flight exclusions); the aggregate contains 5,010 rows, 79 exact rows, and zero mechanically admitted rows across 294 route summaries. Every exact row therefore has a visible reason for rejection or promotion status rather than being silently discarded. The aggregate report is registered separately as an audit artifact, never as a construction family.

5 Reader-facing evaluation

No candidate is called readable by the automatic diagnostics. The release package is gated until a mechanically admitted survivor exists. The planned study uses intact prose and controls within five normalized letters, randomized blinded order, opaque item identifiers with a separate key, and a reproducible rater package. The builder requires structured provenance with immutable generator/source hashes and explicit false flags for copied or borrowed text and shortcut symmetries; it also regenerates controls until their normalized tapes are changed and non-palindromic. Readers rate Englishness, coherence, and grammaticality independently; two blinded coders adjudicate free-form paraphrases. The analysis is preregistered before unblinding, with the candidate admitted only if it beats the controls on the prespecified reader criteria; agreement will be reported with a preregistered reliability statistic such as Krippendorff’s alpha (Krippendorff, 2011). The current manuscript reports no human responses and makes no readability claim.

6 Limitations and next construction

The present snapshot has not yet produced a $\geq$ 100-letter candidate that passes all gates. That is a development status, not the intended contribution. The next run will change the construction state rather than repeat a sweep: the seam lane will use typed boundary resegmentation and then a fresh non-palindromic subject/verb/object frame while preserving agreement; the scene lanes will solve

an object/attachment slot against the live tape; and the LM lane will propose new clause boundaries on held-out vocabulary. Each resulting surface will be rendered and audited before a reader-facing test. Once a survivor appears, the reader package and paper tables will be regenerated from its immutable provenance record.

References

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- Alexandre Papadopoulos, Pierre Roy, Jean-Charles Régin, and François Pachet. 2015. [Generating all possible palindromes from ngram corpora](#). In *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2015)*, pages 2489–2495, Buenos Aires, Argentina.

A Reproduction checklist

Run the ten lane drivers, then the candidate-readability audit and novelty validator scripts in `tools/` and `experiments/`. The former recomputes exactness and mechanical gates; the latter checks unique signatures and required repair artifacts. The committed JSON files under `runs/` contain the rendered text, normalized lengths, hashes, provenance, and next repair for every row. A reader package is generated only after the mechanical admission count becomes nonzero.